Dr Sara-Jayne Thursby

(E): sjthursby94@gmail.com

Education:

Oct. 2016 – June. 2020: PhD Researcher - Genomic Medicine Research Group. Ulster University.

Thesis: "Bioinformatic analysis of epigenetic effects, particularly changes in DNA methylation, following different interventions."

Sept. 2012 – June 2016: University of Ulster, Coleraine (First Class Honours).

BSc. (Hons) Biology with Diploma in International Academic Studies (DIAS).

Dissertation: "Bioinformatic analysis of changes in the genome in response to manipulation of the levels of key epigenetic regulators."

Positions & Scientific Appointments:

Dec 2020 - Present:	Postdoctoral Researcher, Johns Hopkins University
June 2020 – April 2021:	Postdoctoral Researcher, Ulster University/Public Health Authority of Northern Ireland
Oct 2016 – June 2019:	Postgraduate Teacher, Ulster University

Skills Profile:

- Responsible for bioinformatics analysis of next generation sequencing and array-based datasets using:
  Cloud-computing environments
  R Statistical Platform,
  Python
- Facilitates teaching and supervision of undergraduate, PhD students and Postdoctoral Fellows unfamiliar with bioinformatics
- Streamlined data-processing of array and NGS analyses
- Excellent at working as part of a large research group and independently
- Experience in Next Generation Sequencing analysis including variant calling, RNA-seq, WGBS, ChIP-seq, Single Cell multiome
- Fluent in the R programming language.
- Experienced in Unix, Python, Java, HTML, PHP, MySQL and AWS.
- Excellent understanding of bioinformatics and proteomic data manipulation tools such as, Galaxy, UCSC and NCBI genome browsers, GATK, BWA, PICARD
- Co-creation of new tools and documentation for www.usegalaxy.org

Publications:

- **Thursby SJ**, Jin Z, Blum J, Gurau A, Noë M, Scharpf RB, Velculescu VE, Cope L, Brock M, Baylin S, Pisanic T, Easwaran H. Epigenetic Instability Based Metrics in Cell-Free DNA for Early Cancer Detection. Clin Cancer Res. 2026 Jan 27. doi: 10.1158/1078-0432.CCR-25-3384. Epub ahead of print. PMID: 41591979.

- Yang L, Tu L, Bisht S, Mao Y, Petkovich D, **Thursby SJ**, Liang J, Patel N, Yen RC, Largent T, Zahnow C, Brock M, Gabrielson K, Salimian KJ, Baylin SB, Easwaran H. Tissue-location-specific transcription programs drive tumor dependencies in colon cancer. (2024). Nature Communications.

- Downs BM, **Thursby SJ**, Cope L. Detecting aberrant DNA methylation in Illumina DNA methylation arrays: a toolbox and recommendations for its use. (2023). Epigenetics.
- Bhandari YR, Krishna V, Powers R, Parmar S, **Thursby SJ**, Gupta E, Kulak O, Gokare P, Reumers J, Van Wesenbeeck L, Bachman KE, Baylin SB, Easwaran H. Transcription factor

expression repertoire basis for epigenetic and transcriptional subtypes of colorectal cancers. (2023). Proc Natl Acad Sci U S A.

- Irwin RE, Scullion C, **Thursby SJ**, Sun M, Thakur A, Hilman L, Callaghan B, Thompson PD, McKenna DJ, Rothbart SB, Xu G, Walsh CP. The UHRF1 protein is a key regulator of retrotransposable elements and innate immune response to viral RNA in human cells. Epigenetics. 2023 Dec;18(1):2216005. doi: 10.1080/15592294.2023.2216005. PMID: 37246786; PMCID: PMC10228402.
- Ondičová M, Irwin RE, **Thursby SJ**, Hilman L, Caffrey A, Cassidy T, McLaughlin M, Lees-Murdock DJ, Ward M, Murphy M, Lamers Y, Pentieva K, McNulty H, Walsh CP. Folic acid intervention during pregnancy alters DNA methylation, affecting neural target genes through two distinct mechanisms. Clin Epigenetics. 2022 May 16;14(1):63. doi: 10.1186/s13148-022-01282-y. PMID: 35578268; PMCID: PMC9112484.
- **Thursby, S.J.**, Lobo, D.K., Pentieva, K., Zhang, S.D., Irwin, R.E., Walsh, C.P. CandiMeth: Powerful yet simple visualization and quantification of DNA methylation at candidate genes. (2020). Gigascience.
- Amenyah SD, Hughes CF, Ward M, Rosborough S, Deane J, **Thursby SJ**, Walsh CP, Kok DE, Strain JJ, McNulty H, Lees-Murdock DJ. Influence of nutrients involved in one-carbon metabolism on DNA methylation in adults-a systematic review and meta-analysis. Nutr Rev. 2020 Aug 1;78(8):647-666. doi: 10.1093/nutrit/nuz094. PMID: 31977026.
- Lapsley CR, Irwin R, McLafferty M, **Thursby SJ**, O'Neill SM, Bjourson AJ, Walsh CP, Murray EK. Methylome profiling of young adults with depression supports a link with immune response and psoriasis. Clin Epigenetics. 2020 Jun 15;12(1):85. doi: 10.1186/s13148-020-00877-7. PMID: 32539844; PMCID: PMC7477873.
- Irwin, R. E., **Thursby, S.J.**, Ondičová, M., Pentieva, K., McNulty, H., Richmond, R. C., … Walsh, C. P. (2019) A randomized controlled trial of folic acid intervention in pregnancy highlights a putative methylation-regulated control element at ZFP57. (2019). Clinical Epigenetics.
- O'Neill KM, Irwin RE, Mackin SJ, **Thursby SJ**, Thakur A, Bertens C, Masala L, Loughery JEP, McArt DG, Walsh CP. Depletion of DNMT1 in differentiated human cells highlights key classes of sensitive genes and an interplay with polycomb repression. Epigenetics Chromatin. 2018 Mar 29;11(1):12. doi: 10.1186/s13072-018-0182-4. PMID: 29598829; PMCID: PMC5875016.

Research Grants:

- AACR General Meeting-Barbara Campbell Creighton Scholar-in-Training Award ($1500)
- Johns Hopkins Office for Cancer Research Education Training and Education Coordination Travel Award ($750)
- AACR Scholar in Training Award ($500)
- Galaxy Community Fellowship Award (£1600)
- Genetics Society Training Grant (£1000)
- Cancer Translational Research Group (£1000) – for the Biomed Sciences Postgrad Society
- Doctoral Researcher Initiative (£500) – for the Biomed Sciences Postgrad Society
- 'WhyR' Conference Grant (£450)

Oral/Poster Presentations at Conferences:

- Oral presentation: AACR General Meeting (2026)
- Oral & Poster Presentation: AACR DNA Methylation, Clonal Hematopoiesis and Cancer Conference (2025)
- Poster Presentation: American Association For Cancer Research (2024)
- Oral Presentation: Epigenetics in Health and Disease Workshop (2019)
- Oral & Poster Presentation: Galaxy Community and Open-Source Bioinformatics Conference (2018)
- Poster Presentation: European Alliance for Personalized Medicine (2018)

Extracurricular:

- Instructor NIH FEAS Epigenetics (BIOL 039)
- Nominated for 2020 Postgraduate Student of the Year
- International Leadership & Management Award completed to Level 5
- Peer Reviewer for Genomics, Epigenetics, Scientific Reports

References:

Dr. Hariharan Easwaran, Sidney Kimmel Comprehensive Cancer Center, The Johns Hopkins University School of Medicine, Baltimore, 21205
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Prof. Stephen Baylin, Sidney Kimmel Comprehensive Cancer Center, The Johns Hopkins University School of Medicine, Baltimore, 21205
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Dr. Rachelle Irwin, Genomic Medicine Research Group, Ulster University, Coleraine, BT52 1SA
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